

Builder Basics — The Sloped Roof (Extension · Beginner)

Topic: Variables · **Course:** Builder Basics · **Type:** Extension Activity · **Level:** Beginner · **Time:** about 30 minutes

✚ This is a bonus challenge, not a weekly lesson. Try it once you are happy building with loops.

1 · Predict

Open the **Sloped Roof** world but do not run it yet. Here is what the program does.

It builds a roof out of stairs, one square **ring** at a time. A ring is a square outline — like the edge of a picture frame. The program builds the first ring, then builds another ring **on top** of it, then another, then another. Four rings in total, stacked up.

Here is the clever part: each ring is **shorter** than the one below it. A number called `roof_len` decides how long each side of a ring is. It starts at 7. After every ring it loses 2. So the rings get smaller and higher as they go up, and they step up toward a single block at the very top — the peak.

What shape does this make? Circle one: flat square · stepped pyramid roof · single line

Why is each ring smaller than the ring below it?

2 · Watch the Variable Change

12

34

A **variable** is a name that holds a number. The special thing about a variable is that the number can **change** while the program runs.

In this program the variable is called `roof_len` . It holds how long each side of the ring is. It starts at 7, and after each ring it loses 2.

Fill in the table. The first one is done for you.

Layer	<code>roof_len</code> =
Layer 1 (bottom)	7
Layer 2	___
Layer 3	___
Layer 4 (top)	___

The widest ring is at the (circle one): bottom · top

3 · Think It Through

No building yet — just thinking.

The variable `roof_len` loses 2 after each ring. Imagine it did NOT change and stayed 7 the whole time. What would the roof look like then?

Which number would you change to make the roof start out wider at the bottom?

4 · Build It 📷

Now run it for real.

1. Open the **Sloped Roof** world.
2. Type `slope` in the Minecraft chat and press enter to run the program.
3. Watch the rings build up, getting smaller each time, until they reach the peak.

Then change the starting number. Make `roof_len` start at **9** instead of 7, run it again, and see how the roof changes.

Send a photo or video of your roof, then explain it. Use these starters — write 2 or 3 sentences.

My variable `roof_len` started at ... and lost 2 each time, so ...

The roof steps up to a point because the variable ...

When I changed the starting number to 9, the roof ...

Submit ✓

Send this worksheet + a photo or video of your sloped roof to teacher on KakaoTalk.